

AI5707 Pattern Recognition and Machine Learning

Assignment 5: Maximum likelihood parameter estimation

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1. Introduction

The assignment involves fitting Gaussian models to labelled data using maximum likelihood and then using the fitted models to classify points that were held back. For a Gaussian class the maximum likelihood answer is simple, the mean is the average of the training points and the covariance is the average of the outer products of their deviations from that mean.

2. Experiments performed

2.1 Setup

For the synthetic data I drew 10000 points for each class from a Gaussian with means (0, 0), (3, 3) and (6, 0). Standard normal pairs came from the Marsaglia polar method, and I turned them into correlated points by multiplying with the Cholesky factor of the covariance and adding the mean. Each class was then shuffled and split 80/20, giving 8000 training and 2000 test points per class, so 24000 training points and 6000 test points overall. The covariances used are listed in the table below.

Problem	Class 1	Class 2	Class 3
Spherical covariance, identical across classes	[[2, 0], [0, 2]]	[[2, 0], [0, 2]]	[[2, 0], [0, 2]]
Spherical covariance, different across classes	[[1, 0], [0, 1]]	[[2, 0], [0, 2]]	[[3, 0], [0, 3]]
Diagonal covariance, identical across classes	[[3, 0], [0, 1]]	[[3, 0], [0, 1]]	[[3, 0], [0, 1]]
Diagonal covariance, different across classes	[[1, 0], [0, 3]]	[[3, 0], [0, 1]]	[[0.5, 0], [0, 2]]
Full covariance, identical across classes	[[3, 1.2], [1.2, 1.5]]	[[3, 1.2], [1.2, 1.5]]	[[3, 1.2], [1.2, 1.5]]
Full covariance, different across classes	[[2, 1], [1, 2]]	[[2, -1.2], [-1.2, 2]]	[[3, 0.5], [0.5, 1]]

Table 1. True covariance matrices used to generate each synthetic data set.

Estimation follows the structure of the data. For spherical covariance I took the mean of the two diagonal entries of the sample covariance as the single variance, for diagonal covariance I kept only the diagonal, and for full covariance I kept everything. When the classes share a covariance I pooled the three class scatter matrices, weighted by class size, before applying that structure. Priors are the class fractions in the training set, which came out as one third each. A test point goes to the class with the largest value of log prior minus half the log determinant minus half the Mahalanobis distance to the class mean. The boundaries are drawn by classifying a 250 by 250 grid, and the ROC curves treat each class against the other two, using the posterior probability of that class as the score.

2.2 Spherical covariance, identical across classes

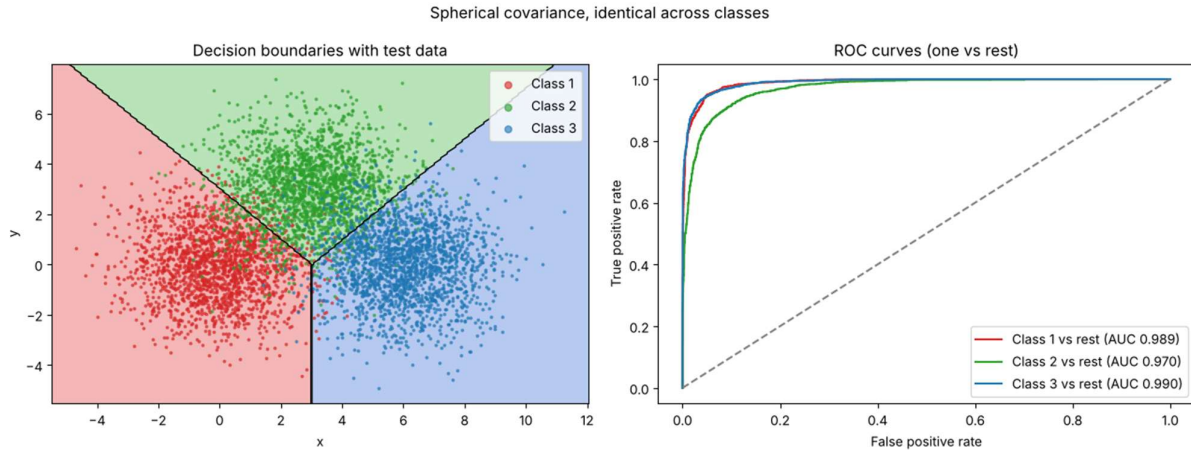


Figure 1. Spherical covariance, identical across classes: decision boundaries with the 6000 test points (left) and ROC curves (right).

True \ Predicted	Class 1	Class 2	Class 3	Total
Class 1	1852	122	26	2000
Class 2	146	1737	117	2000
Class 3	18	129	1853	2000

Table 2. Confusion matrix on the test set, accuracy 90.70%.

With a common spherical covariance the estimates are very close to the truth: the variance comes out as 1.99 against 2, and the means are within 0.04 of the true values. The boundaries are straight lines, and since the priors are equal and the variance is shared they sit halfway between the class means. Accuracy on the test set is 90.70%. Almost all the mistakes are between neighbouring classes, 1 and 2 (122 and 146 points) and 2 and 3 (117 and 129 points), while classes 1 and 3 are hardly mixed up (26 and 18 points). Class 2 has the lowest AUC at 0.970 because it has a neighbour on both sides.

2.3 Spherical covariance, different across classes

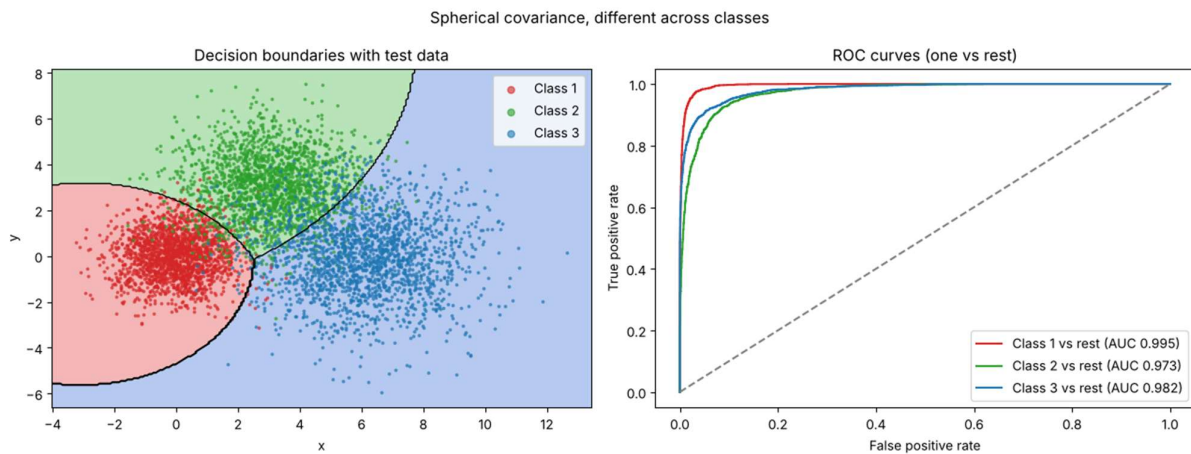


Figure 2. Spherical covariance, different across classes: decision boundaries with the 6000 test points (left) and ROC curves (right).

True \ Predicted	Class 1	Class 2	Class 3	Total
Class 1	1926	60	14	2000
Class 2	87	1768	145	2000
Class 3	26	184	1790	2000

Table 3. Confusion matrix on the test set, accuracy 91.40%.

Once the variances differ the boundaries bend: the tight class 1 gets a rounded region of its own, and the very spread out class 3 takes over the surrounding space. Accuracy is 91.40%. Class 1 is the easiest to pick out (AUC 0.995) because it is compact, whereas classes 2 and 3 overlap more, partly because class 3 is wide and reaches into class 2.

2.4 Diagonal covariance, identical across classes

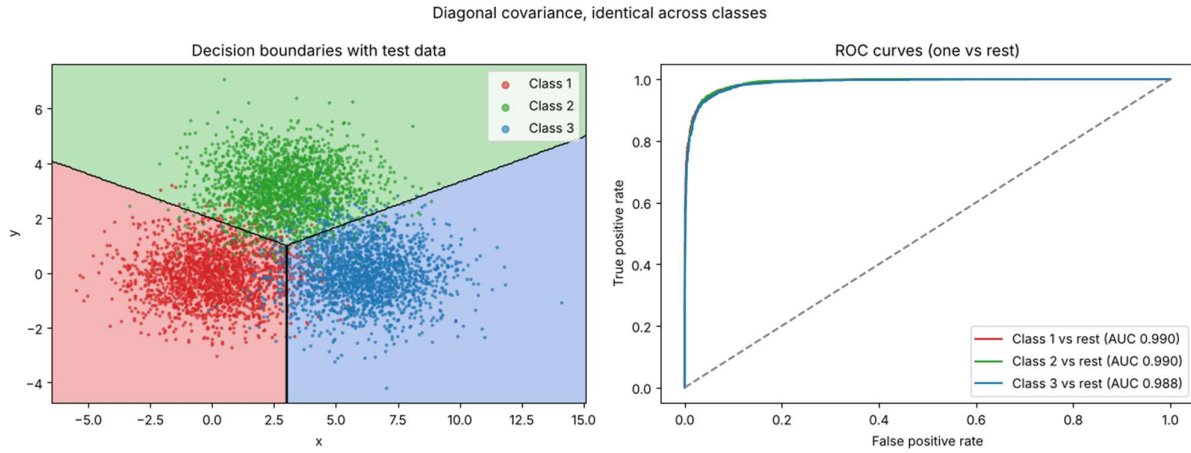


Figure 3. Diagonal covariance, identical across classes: decision boundaries with the 6000 test points (left) and ROC curves (right).

True \ Predicted	Class 1	Class 2	Class 3	Total
Class 1	1848	73	79	2000
Class 2	65	1863	72	2000
Class 3	74	82	1844	2000

Table 4. Confusion matrix on the test set, accuracy 92.58%.

The shared diagonal covariance was found to be $\text{diag}(3.023, 0.998)$ against $\text{diag}(3, 1)$. The classes are stretched along x , and the boundaries are again straight lines, now tilted because the y direction is more informative than the x direction. This gave the best accuracy of the six problems, 92.58%, and the three AUC values are almost the same (0.990, 0.990 and 0.988). The errors are spread fairly evenly over the off-diagonal cells.

2.5 Diagonal covariance, different across classes

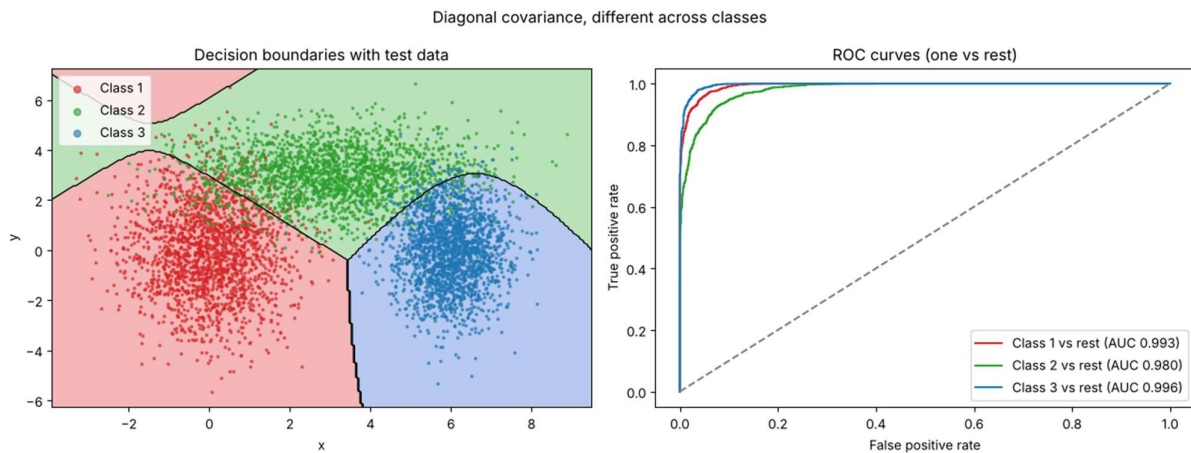


Figure 4. Diagonal covariance, different across classes: decision boundaries with the 6000 test points (left) and ROC curves (right).

True \ Predicted	Class 1	Class 2	Class 3	Total
Class 1	1849	151	0	2000
Class 2	121	1778	101	2000
Class 3	0	81	1919	2000

Table 5. Confusion matrix on the test set, accuracy 92.43%.

All the estimated variances are within 5% of the true ones. The largest gap is the y variance of class 1, which is 2.903 against 3. The boundaries are curved and the plot shows a few regions appearing far from any data, for example class 1 at the top left corner. Accuracy is 92.43%, and classes 1 and 3 are never confused with each other here (0 points in both cells), since they are well separated along x.

2.6 Full covariance, identical across classes

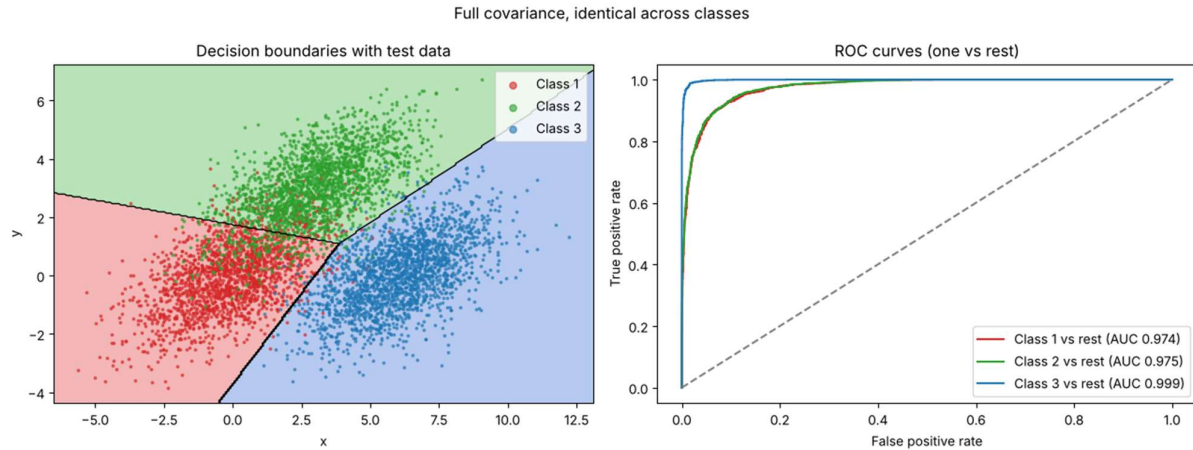


Figure 5. Full covariance, identical across classes: decision boundaries with the 6000 test points (left) and ROC curves (right).

True \ Predicted	Class 1	Class 2	Class 3	Total
Class 1	1743	224	33	2000
Class 2	196	1778	26	2000
Class 3	21	18	1961	2000

Table 6. Confusion matrix on the test set, accuracy 91.37%.

The pooled full covariance is $[[3.016, 1.208], [1.208, 1.506]]$ against $[[3, 1.2], [1.2, 1.5]]$. The boundaries are straight lines. Accuracy is 91.37%. In this problem classes 1 and 2 are confused much more than the others (224 and 196 points), and their AUC values are both about 0.975 while class 3 reaches 0.999. The reason is that the long axis of the shared covariance points roughly along the line joining the means of classes 1 and 2. The Mahalanobis distance between them is only about 2.5, compared with about 4.2 and 4.5 for the pairs involving class 3, so 1 and 2 smear into each other.

2.7 Full covariance, different across classes

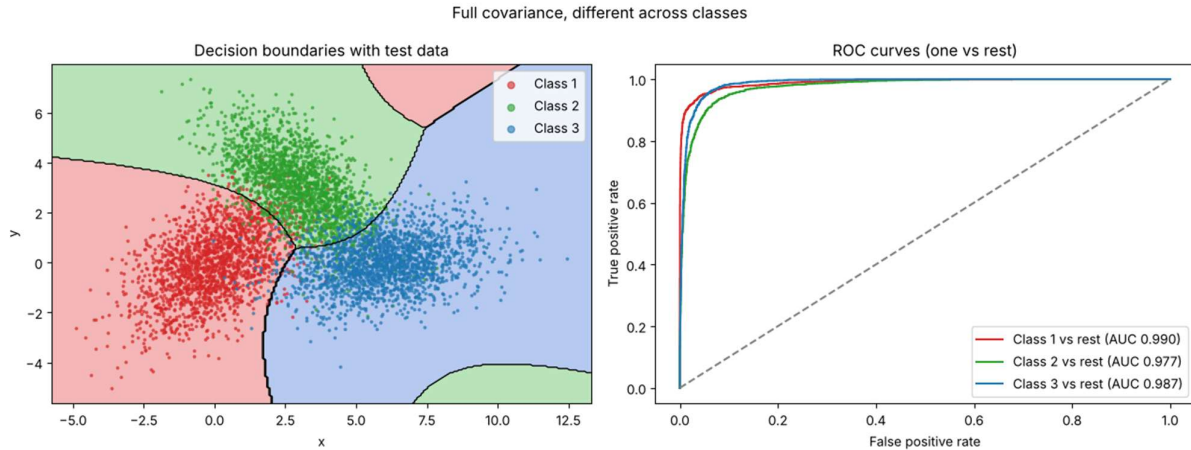


Figure 6. Full covariance, different across classes: decision boundaries with the 6000 test points (left) and ROC curves (right).

True \ Predicted	Class 1	Class 2	Class 3	Total
Class 1	1841	132	27	2000
Class 2	50	1799	151	2000
Class 3	43	87	1870	2000

Table 7. Confusion matrix on the test set, accuracy 91.83%.

This is the most general case and the estimates are still good: the correlations of +0.5, -0.6 and +0.3 in the true covariances show up as 0.50, -0.60 and +0.31 in the estimates. The boundaries are curved, and small regions of one class show up at the edges of the plot, far from where its data lies. Accuracy is 91.83%. The errors are again between neighbouring classes, mostly class 2 going to class 3 (151) and class 1 going to class 2 (132). Class 2 has the lowest AUC (0.977).

2.8 Real data

There were 2 files given, one with 300 points and one with 1050, each given as x, y, class label. I put them together to get 1350 points, 450 per class, and split every class into 360 for training and 90 for testing. I used a full covariance matrix for each class, since the classes clearly have different spreads, and estimated everything with maximum likelihood. The estimated means are (407, 1208), (411, 1919) and (570, 1431), and the estimated covariances are listed below.

Class	Mean	Covariance
1	(406.7, 1208.2)	[[19021, 33535], [33535, 153869]]
2	(411.0, 1919.2)	[[12889, -8713], [-8713, 47721]]
3	(570.2, 1431.4)	[[18665, -1670], [-1670, 49862]]

Table 8. Maximum likelihood estimates for the real data, trained on 1080 points.

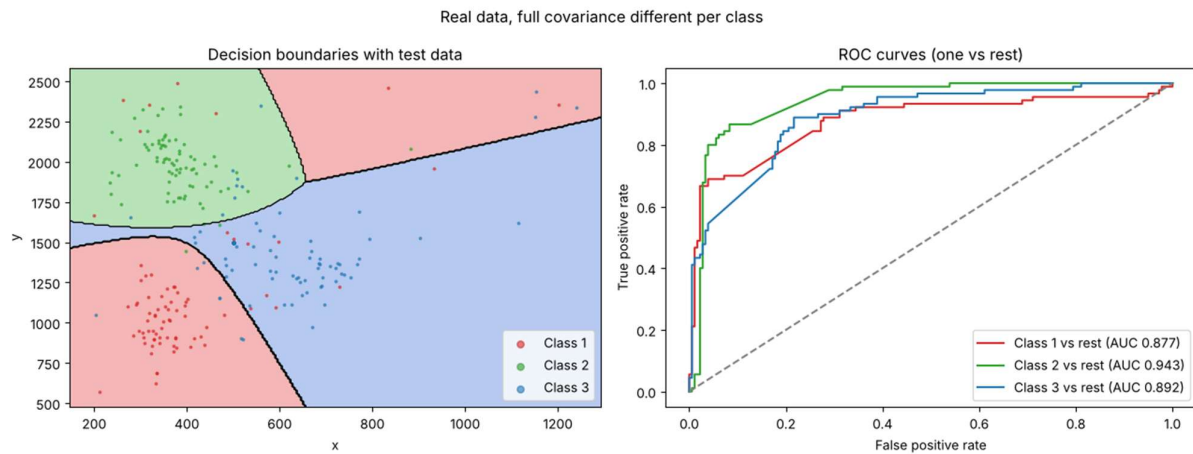


Figure 7. Real data: decision boundaries with the 270 test points (left) and ROC curves (right).

True \ Predicted	Class 1	Class 2	Class 3	Total
Class 1	62	6	22	90
Class 2	2	77	11	90
Class 3	10	9	71	90

Table 9. Confusion matrix on the 270 test points, accuracy 77.78%.

Accuracy is 77.78%, clearly lower than on the synthetic data. Class 2 sits by itself in the upper left and is the easiest, with an AUC of 0.943. Classes 1 and 3 overlap in the middle of the plot, which is where most mistakes are (22 class 1 points went to class 3 and 10 class 3 points went to class 1). Class 1 has a very large spread in y (variance about 154000) and a correlation of roughly 0.62 between x and y, so its fitted ellipse is long and tilted. That ellipse reaches the top right corner, which is why a few far away test points of class 1 are still labelled correctly there. The ROC curves are rough steps rather than smooth curves because there are only 90 test points for each class, and a difference of a couple of points in accuracy on 270 test points is within noise.

2.9 All problems together

Problem	Accuracy	AUC 1	AUC 2	AUC 3
Spherical covariance, identical across classes	90.70%	0.989	0.970	0.990
Spherical covariance, different across classes	91.40%	0.995	0.973	0.982
Diagonal covariance, identical across classes	92.58%	0.990	0.990	0.988
Diagonal covariance, different across classes	92.43%	0.993	0.980	0.996
Full covariance, identical across classes	91.37%	0.974	0.975	0.999
Full covariance, different across classes	91.83%	0.990	0.977	0.987
Real data, full covariance different per class	77.78%	0.877	0.943	0.892

Table 10. Test accuracy and one-versus-rest AUC for every classification problem.

3. Inference

Maximum likelihood estimation works very well when the model matches the data. With 8000 training points per class the estimated means were within about 0.05 of the truth and the covariance entries within a few percent. The covariance estimate divides by n rather than $n - 1$, which makes it slightly biased, but with this many points the difference is negligible.

The covariance structure controls the shape of the boundary. Shared covariance gives straight lines, because the quadratic terms cancel between classes, and different covariances give curved boundaries. The curved boundaries can also create regions far from the data, which is harmless for classification here but shows that quadratic boundaries should not be trusted outside the range of the training data.

On the real data the same model reaches 77.8% accuracy and AUC values between 0.88 and 0.94. The gap to the synthetic results comes from the classes overlapping more and from the classes not being exactly Gaussian, since class 1 in particular is stretched and has a long tail. Maximum likelihood still gives a reasonable classifier, but the result is only as good as the Gaussian assumption, and with 90 test points per class the numbers carry more uncertainty than the synthetic ones.